

“You Do It” Solution

Lecture 7B: Confidence Interval Estimation

Prepared as a classroom-ready detailed solution set for ECOM5005

Exercise 1: Confidence Interval for Perth Apartment Rent

Question

You want to rent an unfurnished one-bedroom apartment in Perth, WA next year. The mean monthly rent for a random sample of 60 apartments advertised on the Real Estate website is AUD \$1,000. Assume a population standard deviation of AUD \$200.

(a) Construct a 95% confidence interval.

(b) To what population of apartments can you appropriately infer from your sample in part (a)?

Detailed Solution

Step 1: Choose the correct confidence-interval formula.

The unknown parameter is the population mean monthly rent μ . From the sample,

$$\bar{X} = 1000, \quad \sigma = 200, \quad n = 60.$$

Because σ is assumed known, use the Z interval for a population mean:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Step 2: Find the 95% critical value.

For a 95% confidence level, $\alpha = 0.05$ and $\alpha/2 = 0.025$, so

$$Z_{\alpha/2} = 1.96.$$

Step 3: Calculate the standard error.

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{200}{\sqrt{60}} = 25.8199.$$

The standard error measures the sample-to-sample variability of $\bar{X}$.

Step 4: Calculate the margin of error.

$$e = Z_{\alpha/2} SE(\bar{X}) = 1.96(25.8199) = 50.6070.$$

Step 5: Construct the interval.

$$\bar{X} \pm e = 1000 \pm 50.6070,$$

so

$$949.3930 \leq \mu \leq 1050.6070.$$

Thus, to the nearest cent,

$$\text{AUD } \$949.39 \leq \mu \leq \text{AUD } \$1,050.61.$$

Part (a) answer: the 95% confidence interval for the population mean monthly rent is approximately **AUD \$949.39 to AUD \$1,050.61**.

Exercise 1: Interpretation and Final Answer

Interpretation of the 95% Confidence Interval

We are **95% confident** that the true mean monthly rent for the population represented by the sampling frame is between

$$\text{AUD } \$949.39 \text{ and AUD } \$1,050.61.$$

The 95% refers to the **long-run performance of the method**: if we repeatedly selected random samples of 60 apartments and constructed intervals in this way, about 95% of those intervals would contain the fixed population mean μ .

Part (b): Appropriate Population for Inference

Inference should be restricted to the population represented by the random-sampling frame. Therefore, the appropriate target population is

unfurnished one-bedroom apartments in Perth, WA represented by the same Real Estate website and relevant listing period.

The interval should not automatically be generalised to furnished apartments, larger apartments, properties outside Perth, listings absent from that website, or a different future market. If the sampled advertisements specifically concern apartments available **next year**, then the inference may be framed for that next-year advertised-rental population; otherwise, the sample alone does not justify a projection to next year's market.

Answer Summary

(a)

$$1000 \pm 1.96 \left(\frac{200}{\sqrt{60}} \right) = 1000 \pm 50.6070,$$

so

$$\text{AUD } \$949.39 \leq \mu \leq \text{AUD } \$1,050.61.$$

(b) Infer only to unfurnished one-bedroom Perth apartments represented by the same Real Estate website sampling frame and time period from which the 60 advertisements were randomly selected.

Final Check

- The sample is stated to be random.
- The population standard deviation is assumed known: $\sigma = \text{AUD } \$200$, so a Z interval is appropriate.
- With $n = 60$, the Central Limit Theorem supports an approximately normal sampling distribution of $\bar{X}$ for most population shapes.
- The interval has the required form: **point estimate** $\pm$ **margin of error** $= 1000 \pm 50.6070$.